Mobile MIMO Radar Imaging via Synergistic Deep Learning and Wave Physics Modeling

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***Abstract*—This article presents a multiple-input multiple-output (MIMO) radar imaging and reconstruction framework for mobile Internet of Things (IoT) platforms, such as intelligent robots and autonomous vehicles. It addresses the limited adaptability of existing model-based imaging methods in dynamic environments, as well as the substantial computational and training burden of end-to-end deep learning methods, by decomposing the radar imaging pipeline into three stages and synergistically combining data-driven perception with physics-based modeling. In particular, a lightweight range–Doppler segmentation network (RDSegNet) is used to robustly detect valid target regions (VTRs), followed by a modified hypothetical phase compensation (MHPC) algorithm for velocity disambiguation and wavenumber-domain approximation (WNDA) reconstruction for high-fidelity global image generation. Faithfully adhering to the first principles of wave physics, the proposed method jointly achieves high-fidelity environment reconstruction with very low computational complexity and can be deployed across diverse waveforms and MIMO topologies, thereby providing a foundation for real-time sensing and decision-making for mobile IoT platforms that operate under tight constraints on computation, memory, and latency. Experimental results on both simulated and real-world data validate its effectiveness.**

***Index Terms*—Environment perception, millimeter-wave (mmWave) radar, multiple-input multiple-output (MIMO) imaging.**

I. Introduction

IN INTERNET of Things (IoT) architectures, the perception layer serves as the foundation that bridges the physical and digital worlds. Accurate and efficient environment perception and reconstruction are crucial for the reliability and intelligence of IoT systems. In mobile IoT platforms such as autonomous vehicles and mobile robots, these capabilities enable key vehicle-to-everything (V2X) and smart city functions, including path planning, simultaneous localization and mapping (SLAM), and autonomous navigation [1], [2], [3], [4], [5]. Millimeter-wave (mmWave) radar stands as a promising alternative to widely used optical onboard perception sensors, such as cameras and LiDAR, due to its all-weather reliability, lighting insensitivity, instantaneous velocity measurement, compact size, and low cost [6], [7], [8], [9]. Among various mmWave radar architectures, time-division multiplexing (TDM) multiple-input multiple-output (MIMO) frequency-modulated continuous-wave (FMCW) radar is dominant for its high performance and cost effectiveness [10], [11]. In practice, these applications often operate in large-scale dynamic scenes (e.g., urban roads, parking lots, and factories), where surrounding objects exhibit motion relative to the platform and their ranges to onboard sensors vary from centimeters to tens of meters. Radar imaging in such scenes faces significant challenges, particularly for TDM-MIMO radar, that hinder deployment on low-cost, resource-constrained IoT platforms.

A. Challenges and Related Work

Several key challenges arise in large-scale dynamic scene radar imaging. First, accurate detection of valid target regions (VTRs) from the background becomes difficult under strong clutter and multipath reflections. Second, irregular and rapid platform motion introduces phase residues that severely blur the image. Precise estimation of the target's relative velocity is required for effective phase compensation. However, although TDM-MIMO radar offers hardware simplicity, it significantly reduces the maximum unambiguous velocity (MUV), resulting in incorrect velocity estimation. Finally, jointly achieving imaging fidelity and computational efficiency remains unresolved. Most current methods employ the far zone (FZ, i.e., Fraunhofer zone) model in polar coordinates for simplicity. However, as shown in Fig. 1(a), this leads to nonuniform spatial resolution. Consequently, targets farther from the radar suffer from severe distortion, since a small angular error is amplified to a large spatial displacement in the far-field due to enlarged mesh cells. On the contrary, a precise near zone (NZ, i.e., Fresnel zone) model in Cartesian coordinates, as shown in Fig. 1(b), improves reconstruction fidelity but incurs heavy

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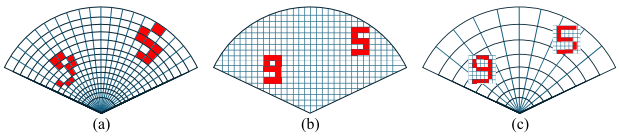


Fig. 1. Reconstruction comparison of shapes "9" and "5" under different coordinate systems. (a) Polar coordinates. (b) Cartesian coordinates. (c) Proposed local Cartesian coordinate.

computational complexity and prohibitive memory demand. In the following, we provide a detailed analysis of the three key limitations along with a review of related work.

1) VTR Detection: The radar detector identifies VTRs from radar images, typically range–Doppler maps (RDMs). The widely used constant false-alarm rate (CFAR) detector [12] is simple, but it exhibits limited robustness to nonuniform clutter. Enhanced variants such as OS-CFAR [13] and censored-CFAR [14] employ adaptive thresholding mechanisms to dynamically accommodate varying noise characteristics, but they are all designed for targets that occupy only a single pixel and are not suitable for extended targets spanning multiple contiguous pixels. To handle this, area-based CFAR [15] redefines the cell under test (CUT) as a fixed rectangular region, but requires manual size tuning, limiting adaptability in dynamic scenes. Recently, data-driven methods offer promising alternatives: DeepRD [16] uses neural networks but is only tested in static chambers; RPDNet [17] combines U-Net and adversarial training with LiDAR-labeled data for dynamic detection, but radial velocity annotation still depends on CFAR. Point cloud (PC)-based approaches [18], [19] risk overfitting due to high-dimensional input. In summary, most existing radar detectors are not suitable for effective VTR detection in high-resolution MIMO radar imaging under dynamic and large-scale scenarios.

2) Velocity Disambiguation: In the TDM-MIMO scheme, the pulse repetition frequency (PRF) decreases inversely with the number of transmitters [20]. When the target's Doppler frequency exceeds half the PRF, the Nyquist sampling criterion is violated, resulting in spectral aliasing and velocity ambiguity [21], [22], which leads to motion compensation errors. Unfortunately, relative velocities of targets often exceed the MUV for fast-moving platforms. Increasing the PRF can mitigate this issue, but inevitably requires reducing the number of transmitters, which in turn degrades spatial resolution. To resolve this, some approaches operate with different PRFs, often implemented using techniques like the Chinese remainder theorem (CRT) [23]. Such techniques, however, require at least two distinct chirp configurations, increasing hardware complexity. Phase-coding methods suffer from limited robustness, with performance degrading as the number of antennas grows or when multiple targets occupy the same RDM cell. The conventional hypothetical phase compensation (HPC) [24] method is straightforward and requires no hardware or waveform modifications. However, its reliance only on peak angular spectra renders it highly sensitive to sidelobes and clutter, resulting in poor robustness.

3) Image Fidelity and Computational Efficiency: In large-scale mmWave imaging on mobile platforms, targets span both FZ and NZ. Under the FZ model, a plane wave approximation enables polar-domain reconstruction where range and angle are decoupled, requiring a regular array for direction-of-arrival (DOA) estimation. To achieve high-fidelity imaging, finer angular grids and superresolution techniques are often used. Subspace-based DOA algorithms such as multiple signal classification (MUSIC) [25] have been proposed to surpass the Rayleigh limit. However, they require prior knowledge of the number of sources and rely on covariance estimation and eigen-decomposition, leading to high computational complexity. Alternatively, DOA estimation can be formulated as a linear inverse problem solved by compressed sensing (CS) [26], [27], [28], which enables angular superresolution. In practice, the coexistence of strong and weak scatterers in dynamic scenes complicates hyperparameter tuning. Shrinkage-based CS methods often suppress low-amplitude components, further reducing robustness to weak targets. More critically, CS-based reconstruction relies on iterative optimization, which incurs higher computational overhead than noniterative methods [29]. For the NZ model reconstruction, wave propagation follows the spherical wavefront assumption, coupling range and angle, requiring joint estimation. While 2D-CS and 2D-MUSIC can be applied, their measurement and covariance matrices grow exponentially with scene size, making the computational cost prohibitive. Recently, deep learning-based methods [30], [31], [32] have demonstrated remarkable imaging performance. However, these end-to-end approaches usually rely on large-scale models and data-intensive training, making them difficult to deploy on edge-computing platforms. Moreover, they often omit Doppler and scattering-intensity information, which are crucial in radar imaging. Back-projection (BP) is the most robust but computationally prohibitive, while the fast range migration algorithm (RMA) offers limited imaging coverage [33], [34]. To address the challenge of cross-zone large-scale scene imaging [35], several imaging algorithms have been proposed [36], [37]; nevertheless, they still rely on FZ polar coordinates. Overall, no unified model has been specifically developed for transitional zone (TZ) reconstruction that can be applied across both FZ and NZ while simultaneously achieving reconstruction accuracy, flexibility, and computational efficiency.

B. Our Contribution

This article addresses the above challenges through the following contributions.

1) A MIMO radar imaging framework for mobile IoT platform sensing is proposed, which synergistically combines data-driven perception with wave physics-based modeling and jointly achieves high imaging fidelity, computational efficiency, and general applicability across diverse radar waveforms and array configurations.
2) A range–Doppler segmentation network (RDSegNet) is designed to enhance VTR detection accuracy and robustness, along with a proposed radial velocity label generation method for automatic annotation.
3) A modified HPC (MHPC) algorithm is developed to improve the robustness of velocity estimation.

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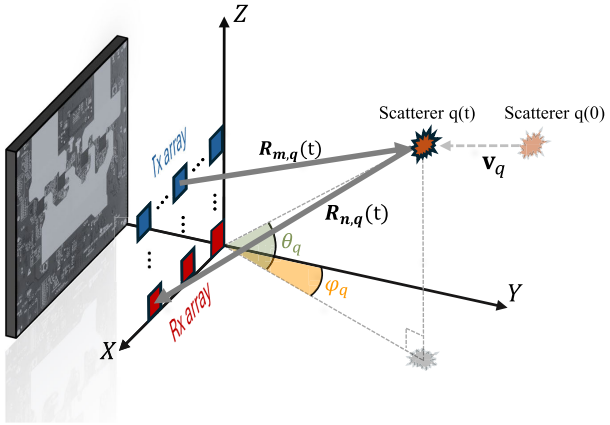


Fig. 2. Geometry model of MIMO radar wave propagation.

4) A local Cartesian coordinate reconstruction TZ model is proposed under wavenumber-domain [i.e., K-space] approximation (WNDA), as shown in Fig. 1(c), reducing both distortion and computational complexity.

The structure of this article is as follows. In Section II, the wave propagation geometry and modeling of the TDM-MIMO FMCW signal are derived. In Section III, the proposed three-stage imaging framework is presented, including VTR detection by RDSegNet, velocity disambiguation by MHPC, and wavenumber-domain imaging by WNDA. In Section IV, performance evaluations on both simulated and real-measured data are provided. In Section V, analysis of the WNDA imaging is given. Finally, Section VI concludes this article.

II. Modeling of TDM-MIMO FMCW Signal

This section introduces the wave propagation model and the modeling of the TDM-MIMO FMCW signal for imaging moving point scatterers. In practice, the scattering by large scatterers can be considered as a superposition of scattering by individual point scatterers. The analytical expression of RDM is derived from first principles of wave physics, which provides a foundation for us to propose imaging algorithms.

A. Geometry Model of MIMO Wave Propagation

For a MIMO radar system with M transmit antennas (Tx) and N receive antennas (Rx), the wave propagation geometry between the antennas and a single moving scatterer is illustrated in Fig. 2. During radar capture, each frame is processed as an independent unit to form a complete radar image. For simplicity and without loss of generality, we focus on a single frame, where the time variable t within the frame time is referenced to $t = 0$ at its start. At $t = 0$, the position and velocity of the qth scatterer are defined as $\mathbf{r}_q = [x_q, y_q, z_q]^T$ and $\mathbf{v}_q = [v_{x,q}, v_{y,q}, v_{z,q}]^T$, where $q = 1, \ldots, Q$ indexes the illuminated scatterers within the imaging scene. The coordinates of the mth Tx and the nth Rx are: $\mathbf{r}_{\mathrm{Tx}}^m = [x_m, y_m, z_m]^T$, $m = 1, \ldots, M$; $\mathbf{r}_{\mathrm{Rx}}^n = [x_n, y_n, z_n]^T$, $n = 1, \ldots, N$. Each scatterer is assumed to move at a constant velocity within one frame capture time. The instantaneous displacement vector between the mth Tx and the qth scatterer at time t is given by

$$\mathbf{R}_{m,q}(t) = \mathbf{R}_{m,q}(0) + \mathbf{v}_q t \tag{1}$$

where $\mathbf{R}_{m,q}(0) = \mathbf{r}_q - \mathbf{r}_{\mathrm{Tx}}^m$ represents the displacement between the qth scatterer and the mth Tx at the reference time. Similarly, the instantaneous displacement between the qth scatterer and the nth Rx at time t can be expressed as

$$\mathbf{R}_{n,q}(t) = \mathbf{R}_{n,q}(0) + \mathbf{v}_q t \tag{2}$$

where $\mathbf{R}_{n,q}(0) = \mathbf{r}_q - \mathbf{r}_{\mathrm{Rx}}^n$ denotes the initial displacement vector between the qth scatterer and the nth Rx. Therefore, the instantaneous wave propagation path length from the mth Tx, reflected by the qth scatterer and received by the nth Rx is written as

$$R_{m,n,q}(t) = \left|\mathbf{R}_{m,q}(0) + \mathbf{v}_q t\right| + \left|\mathbf{R}_{n,q}(0) + \mathbf{v}_q t\right| \tag{3}$$

where $|\cdot|$ denotes the Euclidean norm operation. As derived in Appendix A, the first-order path model can be adopted if the transverse velocities and initial ranges of the scatterers satisfy

$$\frac{v_{\perp,m,q}^2}{R_{m,q}(0)} + \frac{v_{\perp,n,q}^2}{R_{n,q}(0)} \le \frac{\lambda_0}{4T_s^2} \tag{4}$$

where $v_{\perp,m,q}$ and $v_{\perp,n,q}$ are the transverse velocity components orthogonal to the Tx–scatterer and Rx–scatterer lines of sight, respectively; $R_{m,q}(0) = |\mathbf{R}_{m,q}(0)|$ and $R_{n,q}(0) = |\mathbf{R}_{n,q}(0)|$ are the initial ranges; λ_0 is the carrier wavelength; and T_s is the frame time. Therefore, the approximately linear relationship between time and the length of the round-trip wave propagation path is

$$\begin{aligned} R_{m,n,q}(t) &= R_{m,q}(t) + R_{n,q}(t) \\ &\approx R_{m,q}(0) + v_{r,m,q} t + R_{n,q}(0) + v_{r,n,q} t \end{aligned} \tag{5}$$

where $v_{r,m,q}$ and $v_{r,n,q}$ denote the radial velocities with respect to corresponding element. The detailed derivation and analysis are provided in Appendix A.

B. TDM-MIMO FMCW Signal Model

The transmitting waveform of the TDM-MIMO FMCW radar is illustrated in Fig. 3. Each frame time window T_s is divided into N_s slow-time slots. Within each slow-time slot n_s, M chirp signals are sequentially transmitted by M Txs and distinguished at the Rx side based on their dedicated time slots. Each chirp of duration T_c consists of an active pulse interval T_f and an idle time, which ensures sufficient Tx switching and hardware settling time. Therefore, t can be expressed as

$$t = t_f + (m-1)T_c + (n_s - 1)\frac{T_s}{N_s} \tag{6}$$

where $t_f \in [0, T_f]$ denotes the fast time. The chirp pulse $p(t_f)$ is an FMCW signal, which is given as

$$p(t_f) = A \exp\left[j2\pi\left(f_0 t_f + \frac{1}{2}\kappa_f t_f^2\right)\right] \tag{7}$$

where A is the signal amplitude, f_0 is the starting frequency of the chirp, and κ_f denotes the chirp slope. Under the single-scattering assumption [38], the received signal, modeled via

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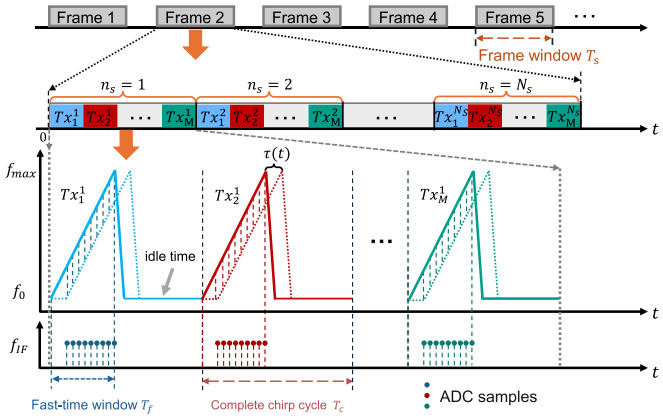


Fig. 3. Waveform of the TDM-MIMO FMCW signal.

the scalar Green's function, is a sampled coherent superposition of waves scattered from all illuminated scatterers, and can be expressed as

$$\tilde{\mathbf{S}}\left(t_f, n_s, m, n\right) = \sum_{q=1}^{Q} \frac{1}{16\pi^2} \cdot \frac{\omega_T \omega_R \varepsilon_q}{R_{m,q} R_{n,q}} \cdot p\left(t_f - \tau_{m,n,q}\right) = \sum_{q=1}^{Q} \frac{A}{16\pi^2} \cdot \frac{\omega_T \omega_R \varepsilon_q}{R_{m,q} R_{n,q}} \cdot \exp\left\{ j2\pi \left[f_0\left(t_f - \tau_{m,n,q}\right) + \frac{1}{2}\kappa_f \left(t_f - \tau_{m,n,q}\right)^2 \right] \right\} \tag{8}$$

where $\tilde{\mathbf{S}}$ denotes the received signal before dechirping, and ω_T and ω_R represent the gain patterns of the Tx and Rx array, respectively. ε_q is the scattering coefficient of the qth scatterer. The wave propagation time delay $\tau_{m,n,q}$ is given by

$$\tau_{m,n,q}(t) \approx \frac{R_{m,n,q}(0) + v_{r,m,q} t + v_{r,n,q} t}{c} \tag{9}$$

where c is the wave propagation speed. After dechirping, the continuous intermediate frequency (IF) signal is obtained as

$$\begin{aligned} \hat{\mathbf{S}}\left(t_f, n_s, m, n\right) &= \tilde{\mathbf{S}}\left(t_f, n_s, m, n\right) \cdot p^*\left(t_f\right) \\ &= \sum_{q=1}^{Q} \frac{A^2}{16\pi^2} \cdot \frac{\omega_T \omega_R \varepsilon_q}{R_{m,q} R_{n,q}} \cdot \exp\left(j\pi\kappa_f \tau_{m,n,q}^2\right) \\ &\quad \cdot \exp\left(-j2\pi\kappa_f \tau_{m,n,q} t_f\right) \cdot \exp\left(-j2\pi f_0 \tau_{m,n,q}\right) \end{aligned} \tag{10}$$

where $\hat{\mathbf{S}}$ denotes the dechirped analog receiving signal and $*$ represents the complex conjugation. Substituting (9) into (10) and omitting negligible phase terms, the discrete IF signal after analog-to-digital conversion (ADC) can be expressed as

$$\begin{aligned} \mathbf{S}\left(n_f, n_s, m, n\right) \approx \sum_{q=1}^{Q} \frac{A^2}{16\pi^2} \cdot \frac{\omega_T \omega_R \varepsilon_q}{R_{m,q} R_{n,q}} &\cdot \exp\left(-j\frac{2\pi f_0 R_{m,n,q}}{c}\right) \\ &\cdot \exp\left(-j\frac{2\pi\kappa_f R_{m,n,q}}{c} \cdot \frac{n_f T_f}{N_f}\right) \\ &\cdot \exp\left(-j\frac{2\pi f_0 v_{r,m,n,q}}{c} \cdot \frac{(n_s - 1) T_s}{N_s}\right) \\ &\cdot \exp\left[-j\frac{2\pi f_0 v_{r,m,n,q}}{c} \cdot (m-1) T_c\right] \\ &\cdot \exp\left[j\pi\kappa_f \left(\frac{R_{m,n,q}}{c}\right)^2\right] \end{aligned} \tag{11}$$

where $\mathbf{S} \in \mathbb{C}^{N_f \times N_s \times M \times N}$, $v_{r,m,n,q} = v_{r,m,q} + v_{r,n,q}$ is the two-way radial velocity, and $n_f = \lfloor t_f N_f / T_f \rfloor$ denotes the fast-time index, with N_f being the total number of ADC samples per chirp and $\lfloor \cdot \rfloor$ being the floor function. The last two exponential terms account for the Doppler phase induced by TDM switching delay (TSD) and the residual video phase (RVP), respectively.

C. Range–Doppler Map

RDMs are generated by applying 2-D fast Fourier transforms (2D-FFT) to $\mathbf{S}$ along n_f and n_s dimensions [39], which yields the following:

$$\begin{aligned} \mathbf{S}_{RDM}\left(f_f, f_s, m, n\right) &= \mathscr{F}_{2D}\left[\mathbf{S}\left(n_f, n_s, m, n\right)\right] \\ &\approx \sum_{q=1}^{Q} \frac{A^2}{16\pi^2} \cdot \frac{\omega_T \omega_R \varepsilon_q}{R_{m,q} R_{n,q}} \cdot \exp\left(-j\frac{2\pi f_0 R_{m,n,q}}{c}\right) \\ &\quad \cdot \exp\left[-j\frac{2\pi f_0 v_{r,m,n,q}}{c} \cdot (m-1) T_c\right] \\ &\quad \cdot \exp\left[j\pi\kappa_f \left(\frac{R_{m,n,q}}{c}\right)^2\right] \\ &\quad \cdot \frac{1 - \exp\left[j2\pi\left(f_{rq} T_f - f_f\right)\right]}{1 - \exp\left(j2\pi \frac{f_{rq} T_f - f_f}{N_f}\right)} \\ &\quad \cdot \frac{1 - \exp\left[j2\pi\left(f_{dq} T_s - f_s\right)\right]}{1 - \exp\left(j2\pi \frac{f_{dq} T_s - f_s}{N_s}\right)} \end{aligned} \tag{12}$$

where f_f and f_s denote the range frequency and Doppler frequency; $\mathscr{F}(\cdot)$ denotes the fast Fourier transform (FFT) operator; $f_{rq} = -\kappa_f R_{m,n,q}/c$; $f_{dq} = -v_{r,m,n,q}/\lambda_0$; and λ_0 is the wavelength corresponding to the carrier frequency f_0. The last two terms in (12) represent the range and velocity ambiguity functions, which together define the RDM. Local peaks in the RDM are selected as candidate targets. However, clutter, multipath reflections, and antenna sidelobes may also cause spurious high responses. Therefore, reliable target detection is essential for high-resolution MIMO radar imaging.

III. Imaging Algorithm Design

In this section, as illustrated in Fig. 4, the proposed imaging method is introduced by decomposing the pipeline into three stages: VTR detection, velocity disambiguation, and wavenumber-domain imaging. Stage A uses the RDSegNet to detect VTR from RDMs. Stage B uses the MHPC to resolve radial velocity ambiguity. Stage C uses the WNDA to reconstruct domains of interest (DOI) subimages and spatially integrate into the global radar image.

A. VTR Detection: RDSegNet

VTR detection in RDM can be regarded as an image segmentation problem, which has been widely explored in

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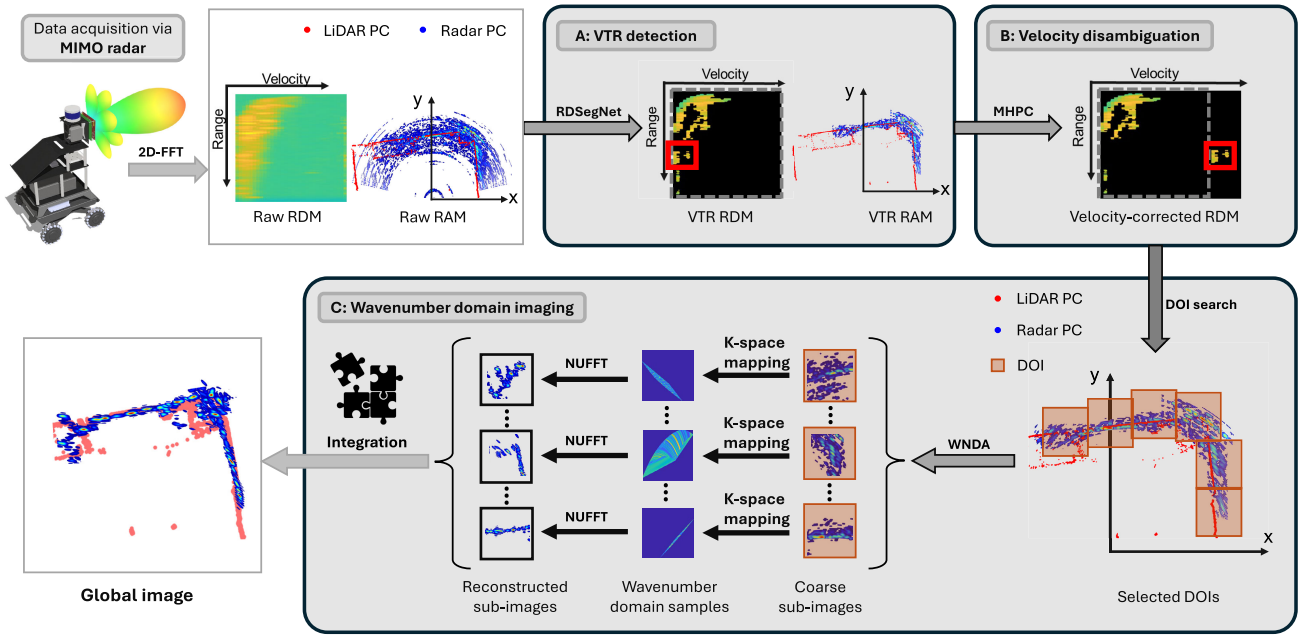


Fig. 4. Schematic of the proposed method. As the radar transmits and captures echo signals during platform movement, 2D-FFT is first applied to the echo data to generate RDMs, which are then processed in three stages. In Stage A, RDSegNet is employed to detect the VTR on raw RDM. In Stage B, for each cell within the VTR, radial velocity disambiguation and velocity correction are performed; the corrected velocities are then used to compensate for the motion-induced phase residue. In Stage C, the VTR is partitioned into multiple DOIs. For each DOI, the phase-compensated echo tensor is mapped into the corresponding wavenumber domain, and a subimage is reconstructed via the NUFFT. Finally, the reconstructed regional subimages from all DOIs are spatially integrated to form a global high-fidelity radar image. The LiDAR PC is used as the GT to evaluate the accuracy of radar image reconstruction. The range-azimuth maps (RAMs) are shown only for Cartesian visualization of the raw and VTR-extracted radar PCs.

the computer vision community using deep learning (e.g., U-Net). RDSegNet is built upon a U-shaped architecture, where a complex-valued convolution module is employed to preserve the algebraic structure of complex-valued radar data inputs. Attention mechanisms are also integrated to enhance interchannel phase feature extraction.

1) LiDAR PC Radial Velocity Generation: Before designing the network, the first challenge lies in generating training datasets. A LiDAR PC, with its superior spatial resolution and reduced clutter sensitivity, is often used as the ground truth (GT) and label for radar imaging evaluation. However, it lacks Doppler capability, and thus radial velocity cannot be directly obtained from synchronized LiDAR data for Doppler-dimension labeling. Fortunately, based on motion model analysis, we propose a method to compute the radial velocity of each LiDAR point.

During dataset collection, LiDAR and radar are mounted on the same platform and operate simultaneously, with LiDAR PC projected onto the "LiDAR RDM" as label annotations. The range of each point is readily calculated from its 3-D Cartesian coordinates (x, y, z). Interestingly, during ego-motion, static targets are observed to exhibit a velocity equal in magnitude but opposite in direction to the platform's velocity $\mathbf{v}_{\text{Plat}}$, as shown in Fig. 5, which can be expressed as

$$\mathbf{v}_1 = \mathbf{v}_2 = \ldots = \mathbf{v}_q = -\mathbf{v}_{\text{Plat}}. \tag{13}$$

For scatterer q, as analyzed in Section II, the velocity observed in the RDM corresponds to the radial component of its true velocity. By decomposing $\mathbf{v}_q$ along the x-, y-, z-axes and

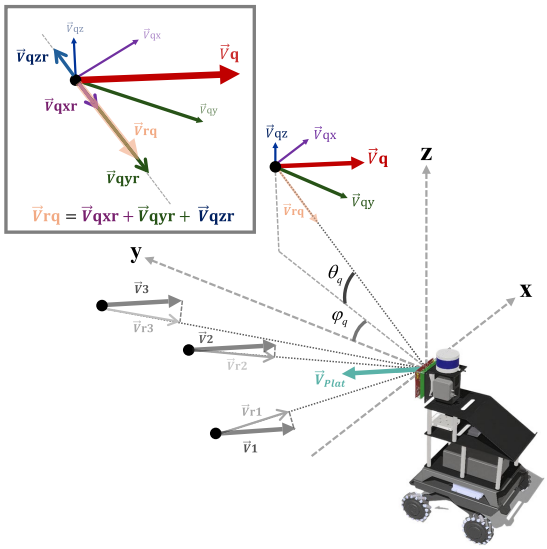


Fig. 5. Velocity model and decomposition for radial velocity generation of LiDAR PC.

applying the distributive property of vector projection

$$\begin{aligned} v_{rq} &= \mathbf{v}_q \cdot \frac{\mathbf{r}_q}{|\mathbf{r}_q|} = \mathbf{v}_{qx} \cdot \frac{\mathbf{r}_q}{|\mathbf{r}_q|} + \mathbf{v}_{qy} \cdot \frac{\mathbf{r}_q}{|\mathbf{r}_q|} + \mathbf{v}_{qz} \cdot \frac{\mathbf{r}_q}{|\mathbf{r}_q|} \\ &= v_{qx} \sin\varphi_q \cos\theta_q + v_{qy} \cos\varphi_q \cos\theta_q + v_{qz} \sin\theta_q \end{aligned} \tag{14}$$

where φ_q and θ_q denote the azimuth and elevation angles of scatterer q, respectively. During dataset acquisition, the platform velocity $\mathbf{v}_{\text{Plat}}$ can be easily obtained using onboard sensors such as an inertial measurement unit (IMU),

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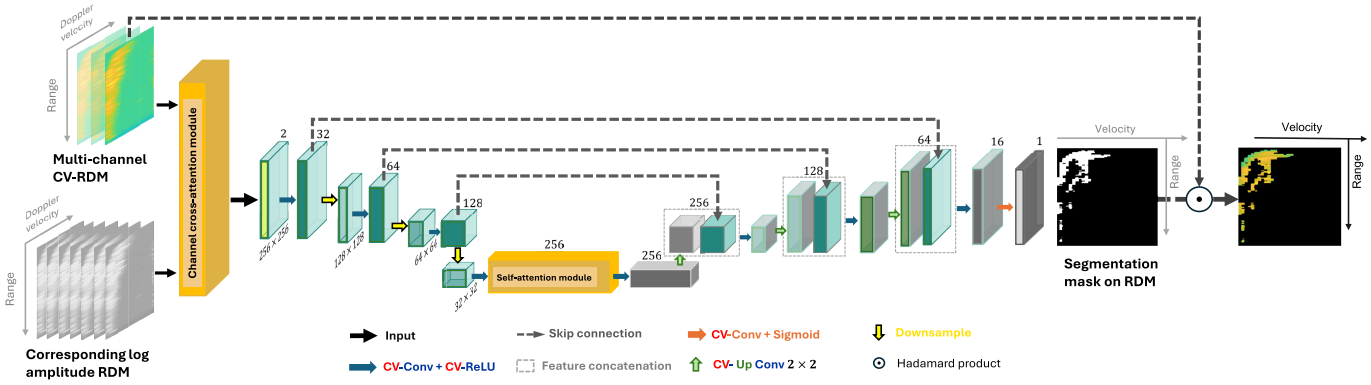


Fig. 6. Architecture of RDSegNet. The RDMs and their corresponding amplitude values from three predefined Tx–Rx pairs are used as inputs. After processing through RDSegNet, a mask is generated, which is then applied to the RDMs of all Tx–Rx pairs via Hadamard product to achieve VTR segmentation and clutter removal.

odometers, or through LiDAR PC registration. Consider all scatterers together, (14) is converted into the matrix format

$$\begin{bmatrix} v_{r1} \\ \vdots \\ v_{rq} \\ \vdots \\ v_{rQ} \end{bmatrix} = \begin{bmatrix} \sin\varphi_1\cos\theta_1 & \cos\varphi_1\cos\theta_1 & \sin\theta_1 \\ \vdots & \vdots & \vdots \\ \sin\varphi_q\cos\theta_q & \cos\varphi_q\cos\theta_q & \sin\theta_q \\ \vdots & \vdots & \vdots \\ \sin\varphi_Q\cos\theta_Q & \cos\varphi_Q\cos\theta_Q & \sin\theta_Q \end{bmatrix} \begin{bmatrix} -v_{\text{Plat},x} \\ -v_{\text{Plat},y} \\ -v_{\text{Plat},z} \end{bmatrix}. \tag{15}$$

Thus, the radial velocity of each point can be calculated using (15). These values are treated as GT and projected onto the synthetic "LiDAR RDM," which serves as supervision for training RDSegNet. It is worth mentioning that (15) remains valid even with platform rotation, as the induced rotational velocity is orthogonal to the radial direction.

2) RDSegNet Architecture: As shown in Fig. 6, RDSegNet adopts a three-stage U-shaped backbone. For high-resolution radar in large-scale environments, VTR detection on RDMs is no longer a simple detection task but a segmentation problem. Unlike natural images with three red-green-blue (RGB) channels, an RDM is complex valued, and each Tx–Rx pair generates one RDM. Thus, an M-Tx N-Rx MIMO array produces $M \times N$ RDMs per frame, which would correspond to a $2MN$-channel input if used directly, leading to prohibitive data volumes and training costs. To balance information preservation and complexity, RDSegNet uses three representative RDMs (e.g., first, 64th, and 86th Tx–Rx pairs in our implementation).

Most RDM magnitudes are similar, as only minor variations in $R_{m,n,q}$ and $v_{r,m,n,q}$ arise from Tx–Rx positions, while phase differences remain significant. Therefore, spatially separated Tx–Rx pairs are selected, since their distinct propagation paths and incidence angles make multipath and clutter patterns more learnable. The logarithmic amplitudes of the three complex RDMs guide the cross-attention [40], while self-attention [41] in the bottleneck provides global context. Each stage employs complex-valued convolution (CV-Conv) [42] kernels that strictly follow complex arithmetic and extract phase information, combined with complex-valued ReLU (CV-ReLU) activation [43]. Finally, considering that effective targets occupy only a small fraction of the RDM, focal [44] and dice [45] losses are used to mitigate class imbalance.

B. Velocity Disambiguation: MHPC

In TDM-MIMO systems, the effective slow-time duration is extended by a factor of M, which reduces the MUV by the same factor

$$v_{\max} = \frac{\lambda_0}{4\,(T_s/N_s)} = \frac{\lambda_0}{4MT_c} \tag{16}$$

where $v_{\max}$ denotes the MUV. As a result, in many scenarios, the target's radial velocity exceeds this limit, causing velocity ambiguity. According to (12), the radial velocity is estimated from the peak position in the RDM. However, due to the periodic nature of the FFT, if the true velocity surpasses $v_{\max}$, it will alias and appear as

$$v_{\text{obs}} = v_{r,\text{true}} - 2\zeta \cdot v_{\max} \tag{17}$$

where v_{obs} is the observed radial velocity, $v_{r,\text{true}}$ is the true radial velocity, and $\zeta \in \mathbb{Z}$ is the unknown velocity ambiguity index. Thus, the problem converts to estimating ζ to recover the true radial velocity from (17).

Fortunately, as shown in (12), the Doppler phase term $\exp[-j\,2\pi f_0 v_{r,m,n,q} \cdot (m-1)T_c/c]$, induced by the TSD, is coupled with the spatial term $\exp(-j\,2\pi f_0 R_{m,n,q}/c)$, resulting in imaging blur. The residue causes defocusing in the angular spectrum. In the HPC method [24], candidate ζ values (e.g., -5 to 5) are tested to estimate the true radial velocity. For each candidate, the corresponding radial velocity is computed according to (17) and used to compensate the corresponding RDM cell before generating the angle spectrum. The candidate yielding the most focused and sharp spectrum is selected. Once the optimal ζ is identified, the TSD-induced phase error is fully corrected, producing a well-focused angular spectrum. In the traditional HPC algorithm, the spectral focusing degree is typically validated only by evaluating the spectral peak power.

To improve the robustness of velocity disambiguation, we propose MHPC, a joint metric for evaluating the spectral sharpness by combining three criteria: peak power, signal-to-noise ratio (SNR), and entropy. The joint metric is defined as

$$\mathcal{J}(\zeta) = \mu_1 \cdot \max P(\zeta) + \mu_2 \cdot \text{SNR}(\zeta) - \mu_3 \cdot \mathcal{H}(\zeta) \tag{18}$$

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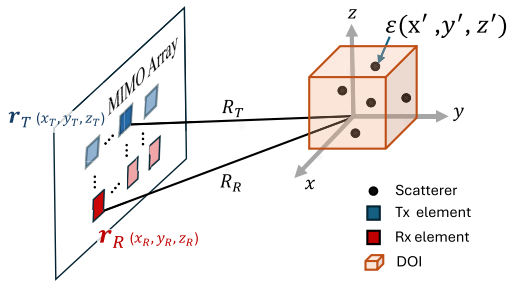


Fig. 7. Illustration of MIMO imaging for a certain chosen DOI.

where μ_1, μ_2, μ_3 are weighting factors used to balance the peak power $\max P(\zeta)$, $\mathrm{SNR}(\zeta)$, and entropy $\mathcal{H}(\zeta)$ of the angular spectrum, given a hypothetical ambiguity index ζ. The optimal estimation of ζ is defined by maximizing the joint criterion

$$\hat{\zeta} = \arg\max_{\zeta} \mathcal{J}(\zeta). \tag{19}$$

For each valid cell in the RDM, the estimated $\hat{\zeta}$ is used to recover the true unambiguous radial velocity, followed by TSD-based phase and RVP compensation. After compensation, FFT-based DOA estimation is performed on the corrected angular spectrum to generate an initial coarse image, from which the DOIs are selected by identifying local peaks.

C. Wavenumber-Domain Imaging: WNDA

After clutter removal outside the VTR and phase compensation, an inverse FFT (IFFT) is applied along the f_f (fast-time frequency) dimension of (12) for each selected $\widetilde{f}_s$ within the VTR, yielding

$$\begin{aligned}\mathring{\mathbf{S}}_{RDM}&\left(n_f, \widetilde{f}_s, m, n\right)\\ &\approx \sum_{q=1}^{Q_{\widetilde{f}_s}} \frac{A^2}{16\pi^2}\cdot\frac{\omega_T\omega_R\varepsilon_q}{R_{m,q}R_{n,q}}\cdot\exp\left(-j2\pi f_0\frac{R_{m,n,q}}{c}\right)\\ &\cdot\exp\left(-j\frac{2\pi\kappa_f R_{m,n,q}}{c}\cdot\frac{n_f T_f}{N_f}\right)\cdot A_{\widetilde{f}_s}\end{aligned} \tag{20}$$

where $\mathring{\mathbf{S}}_{RDM}$ denotes $\mathbf{S}_{RDM}$ after TSD and RVP phase compensation, and $A_{\widetilde{f}_s}$ indicates the velocity ambiguity function introduced amplitude.

1) DOI Subimage Reconstruction: The imaging space containing effective targets is partitioned into multiple DOIs based on the VTR and DOA. For each DOI, independent reconstruction is performed, which can be executed in parallel. Since the number of scatterers is unknown, we assume that each voxel within a DOI may contain a potential scatterer, and the index-based notation is converted into a coordinate-based representation. As illustrated in Fig. 7, the origin is redefined within the DOI. The positions of the Tx and Rx elements are denoted by $\mathbf{r}_T = (x_T, y_T, z_T)$ and $\mathbf{r}_R = (x_R, y_R, z_R)$, respectively, while the scatterer position is given by $\mathbf{r}' = (x', y', z')$. The distances from each Tx and Rx element to the scatterer are respectively rewritten as

$$\begin{aligned}\left|\mathbf{r}_T - \mathbf{r}'\right| &\approx R_T - \mathbf{r}'\cdot\hat{\mathbf{r}}_T\\ \left|\mathbf{r}_R - \mathbf{r}'\right| &\approx R_R - \mathbf{r}'\cdot\hat{\mathbf{r}}_R\end{aligned} \tag{21}$$

where $R_T = |\mathbf{r}_T|$ and $R_R = |\mathbf{r}_R|$ are the Euclidean distances from the defined DOI origin to Tx and Rx, respectively, and $\hat{\mathbf{r}}_T$ and $\hat{\mathbf{r}}_R$ are their corresponding unit direction vectors. The accuracy of the above approximation improves as R_T and R_R increase and $|\mathbf{r}'|$ decreases, indicating that scatterers farther from the antenna element and closer to the origin are more accurately reconstructed [38], [46]. The detailed derivations are provided in Appendix B. For each DOI under a certain $\widetilde{f}_s$, upon the approximation made in (21), the phase-compensated (20) can be written as

$$\begin{aligned}\mathbf{S}_{\widetilde{f}_s DOI}&\left(n_f, \mathbf{r}_T, \mathbf{r}_R\right) \approx \iiint A_{\widetilde{f}_s}\frac{A^2\omega_T\omega_R\,\varepsilon_{\widetilde{f}_s}(\mathbf{r}')}{16\pi^2 R_T R_R}\\ &\cdot\exp\left[-jK\left(R_T + R_R - \mathbf{r}'\cdot(\hat{\mathbf{r}}_T + \hat{\mathbf{r}}_R)\right)\right]d\mathbf{r}'\\ &= \frac{A_{\widetilde{f}_s}A^2\omega_T\omega_R}{16\pi^2 R_T R_R}\cdot\exp\left[-jK\left(R_T + R_R\right)\right]\\ &\cdot\iiint\varepsilon_{\widetilde{f}_s}\left(x', y', z'\right)\\ &\cdot\exp\left(jk_x x'\right)\exp\left(jk_y y'\right)\exp\left(jk_z z'\right)\,dx'dy'dz'\\ &= \frac{\pi A_{\widetilde{f}_s}A^2\omega_T\omega_R}{2R_T R_R}\cdot\exp\left[-jK\left(R_T + R_R\right)\right]\\ &\cdot\mathscr{F}_{3D}^{-1}\left[\varepsilon_{\widetilde{f}_s}\left(x', y', z'\right)\right]\end{aligned} \tag{22}$$

where $K = (2\pi/c)(\kappa_f n_f T_f/N_f + f_0)$ denotes the instantaneous wavenumber at the fast-time index n_f, and the effective wavenumber vector has three components

$$\begin{aligned}k_x &= k_{xt} + k_{xr} = K\left(\frac{x_T}{R_T} + \frac{x_R}{R_R}\right)\\ k_y &= k_{yt} + k_{yr} = K\left(\frac{y_T}{R_T} + \frac{y_R}{R_R}\right)\\ k_z &= k_{zt} + k_{zr} = K\left(\frac{z_T}{R_T} + \frac{z_R}{R_R}\right).\end{aligned} \tag{23}$$

It is evident that the scattering process corresponds to an inverse Fourier transform from the scattering coefficient distribution map in Euclidean space to an approximate wavenumber domain. Intuitively, the WNDA imaging process can thus be realized through a Fourier transform

$$\begin{aligned}\tilde{\boldsymbol{\varepsilon}}_{\widetilde{f}_s}\left(x', y', z'\right) = \mathscr{F}_{3D}\Bigg[&\frac{2\,R_T R_R}{\pi A_{\widetilde{f}_s}A^2\,\omega_T\,\omega_R}\cdot\exp\left(jK\left(R_T + R_R\right)\right)\\ &\cdot\mathbf{S}_{\widetilde{f}_s DOI}\left(n_f, \mathbf{r}_T, \mathbf{r}_R\right)\Bigg].\end{aligned} \tag{24}$$

Due to discrete ADC sampling in the time domain and the spatial distribution of antennas, the corresponding K-space sampling is inherently discrete and nonuniform. Its sampling pattern is jointly determined by the instantaneous wavenumber K, the array configuration, and the DOI origin. To reconstruct the scattering coefficient distribution, nonuniform fast Fourier transform (NUFFT) is employed [47]. The imaging resolution is governed by the coverage of the K-space, and the field-of-view (FOV) is determined by the sampling interval in K-space. For imaging with each DOI as the origin, the model is equivalent to multiple plane waves propagating from the DOI origin to each antenna element and is therefore also called the local plane wave model. It should be noted that objects from other DOIs may also appear, but with severe geometric

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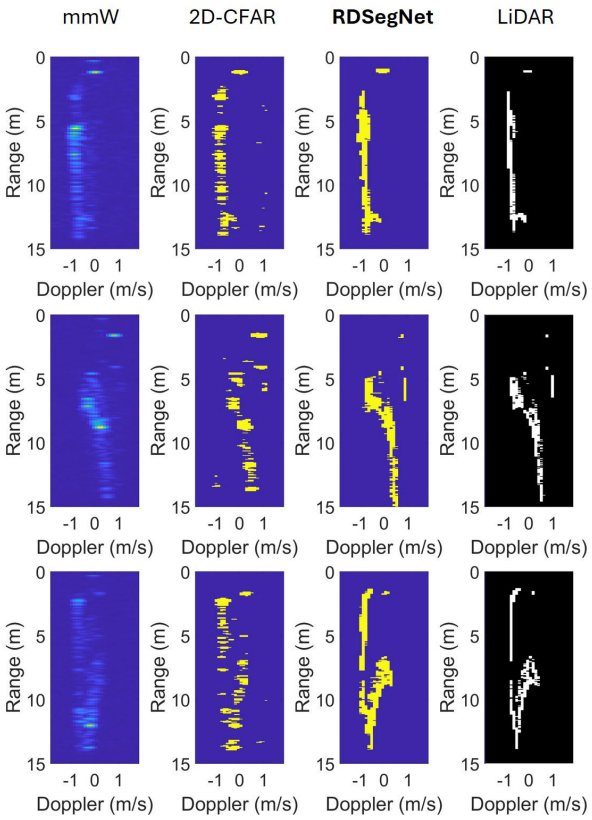


Fig. 8. Visual comparison against 2-D OS-CFAR methods under an indoor environment. The first column shows the original RDMs. The second column presents the VTRs (in yellow) obtained by the OS-CFAR detector. The third column shows the VTRs (in yellow) produced by the proposed RDSegNet detector. The fourth column provides the GT VTRs (in white) derived from LiDAR data.

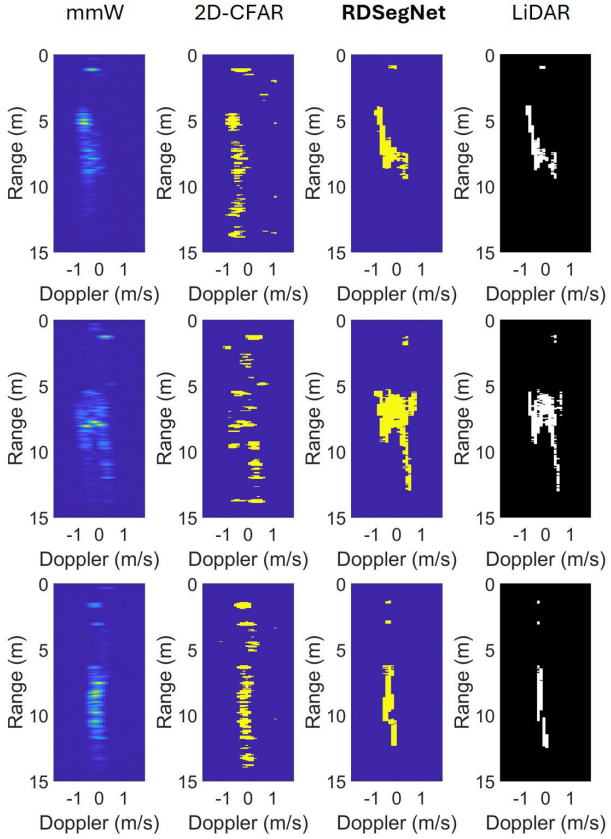


Fig. 9. Visual comparison against 2-D OS-CFAR methods in an outdoor environment. The subfigure layout follows the same arrangement as shown in Fig. 8.

distortions due to the larger error made in (21) as discussed immediately after (21). Therefore, the DOI subimage needs to be cropped to exclude regions outside the predefined fidelity area, which will be illustrated in Section IV-C.

2) Global Image Integration: The global image is generated by integrating all DOI subimages under each selected $\widetilde{f}_s$, and this process is repeated for all valid $\widetilde{f}_s$, meaning that DOIs with different velocities must be incorporated together to form the final global image.

IV. Performance Validation

In this section, systematically designed experiments are conducted to evaluate each module: RDSegNet, MHPC, and WNDA imaging, through quantitative assessments and qualitative visualizations, with comparisons against traditional and state-of-the-art (SOTA) approaches.

A. Evaluation of RDSegNet Detector

Real-world effects such as multipath reflection and clutter are difficult to simulate accurately; therefore, this part of the experiment is conducted solely on measured data.

1) Dataset and Implementation: Alongside the public CoRadar dataset [48] (9000 training, 1000 validation, and 1500 testing frames), a self-collected dataset (1800 training, 200 validation, and 500 testing frames) is also included to enrich scene diversity across corridors, classrooms, laboratories, and an outdoor plaza. LiDAR-radar alignment is ensured via calibration. The radar configuration is listed in Table I (TI-CAS-EVM [49]). RDSegNet is trained and tested on a server equipped with an Advanced Micro Devices, Inc. (AMD) Ryzen 5975WX central processing unit (CPU) at 3.6 GHz and an NVIDIA RTX A5000 graphics processing unit (GPU) with 24 GB of memory. The batch size is set to 32, and the Adam optimizer is used with an initial learning rate of 1×10^{-3}.

TABLE I
Radar Configuration Parameters for Simulation and Real Test

Parameters	Indications /Values
TI-CAS-EVM	12-Tx/16-Rx
2D simulation MIMO	64-Tx/96-Rx
3D simulation MIMO	128-Tx/128-Rx
MIMO scheme	TDM-MIMO
Center frequency	77 GHz
Chirp rate	79 MHz/μs
Sampling rate	8 MHz
Number of samples per pulse	256
Pulse repetition frequency	22.19 kHz
Number of pulses per frame	32
Bandwidth	2.52 GHz

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TABLE II
COMPARISON OF DETECTION PERFORMANCE UNDER DIFFERENT SCENARIOS

Method	PoD (%)			IoU (%)		
	Indoor	Outdoor	Hybrid	Indoor	Outdoor	Hybrid
CA-CFAR	39.4	47.6	41.7	18.0	25.3	24.5
OS-CFAR	40.4	45.5	32.2	20.2	28.1	28.0
Adaptive-CFAR	55.9	61.6	54.0	35.9	49.4	40.2
RDSegNet	**91.7**	**89.8**	**90.5**	**75.6**	**78.0**	**74.7**
RPDNet	75.7	62.5	63.4	50.1	56.7	59.6
RPDNet-LRD	87.4	78.2	74.6	63.4	58.5	61.3

The network is trained for a maximum of 50 epochs. A two-stage learning-rate schedule is adopted: the first five epochs use linear warm-up, and the remaining epochs use cosine annealing to decay the learning rate to a minimum value of 1×10^{-6}. The 10 800 training frames lead to about 16 900 parameter-update iterations, with a total training time of 5.5 h.

2) Quantitative Evaluation: While probability of detection (PoD) is a widely used radar-detection metric, it counts only correct detections, ignores false alarms, and is limited to isolated point targets, thus failing to capture spatially extended objects. By contrast, intersection over union (IoU), a standard semantic-segmentation metric, provides a pixelwise metric reflecting both localization accuracy and spatial coverage. The IoU is defined as

$$\text{IoU} = \frac{\sum\left(\widehat{\boldsymbol{\Theta}} \cap \boldsymbol{\Theta}\right)}{\sum\left(\widehat{\boldsymbol{\Theta}} \cup \boldsymbol{\Theta}\right)} \tag{25}$$

where $\widehat{\boldsymbol{\Theta}}$ denotes the predicted mask, and $\boldsymbol{\Theta}$ represents the GT mask. The operator $\cap$ indicates the elementwise logical conjunction, while $\cup$ denotes the elementwise logical disjunction. As presented in Table II, RDSegNet is compared with two classic CFAR detectors (CA-CFAR and OS-CFAR), one adaptive CFAR, and a SOTA deep learning-based detection network: RPDNet [17]. Our method outperforms all others in both PoD and IoU, indicating not only accurate target detection but also effective suppression of nontarget responses. Despite competitive results, RPDNet lacks supervision along the Doppler dimension and relies on CFAR for velocity detection, leading to lower accuracy than RDSegNet. For a fair network-level comparison, RPDNet was also trained using the same LiDAR-supervised 2-D range–Doppler label (LRD) data, denoted as RPDNet-LRD in Table II. To examine whether RDSegNet overly depends on the large-aperture channel configuration, it was directly evaluated using three closely spaced Tx–Rx channels selected from a single-chip region without retraining, resulting in only 2.23% and 4.07% average decreases in PoD and IoU, respectively. This substitute cross-sensor test reflects its generalization ability.

3) Qualitative Evaluation: Figs. 8 and 9 visually demonstrate that the proposed method significantly outperforms traditional 2-D OS-CFAR, especially in scenarios involving spatially continuous targets. The proposed RDSegNet effectively segments VTR in the RDM, yielding results that closely match the GT provided by LiDAR. In addition, it shows strong robustness in suppressing clutter responses. On the contrary, the CFAR algorithm performs well only for sparse point targets with known maximum size and fails in detecting spatially extended or dense targets.

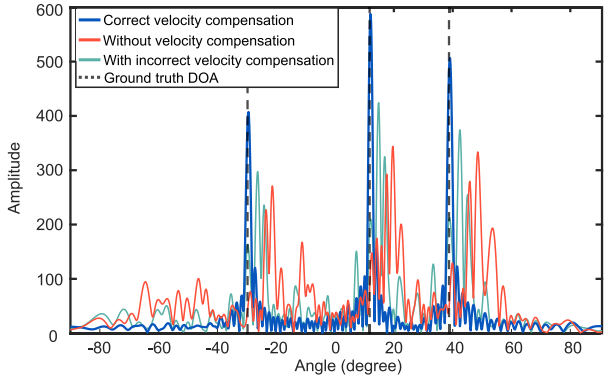


Fig. 10. Impact of velocity compensation on spatial spectrum estimation with ambiguity index = 3.

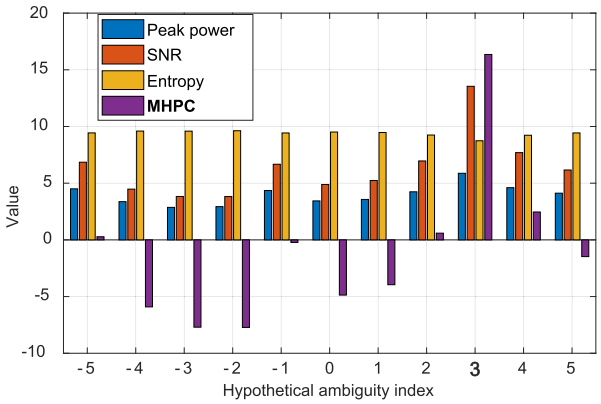


Fig. 11. Comparison of different velocity disambiguation methods: peak power, SNR, entropy, and the proposed MHPC criterion for candidate hypothetical ambiguity indices with GT ambiguity index = 3.

B. Assessment of MHPC Algorithm

To demonstrate the effectiveness of the proposed MHPC algorithm, a simulation is conducted under the configuration presented in Table I, with an unambiguous velocity range of $[-1.7639,\ 1.7639]$m/s. Three targets are placed at the same range, with identical radial velocity of 11.5834 m/s but located at different azimuth angles of $-30°$, $12°$, and $39°$. Due to the velocity spectrum aliasing, the observed radial velocity is 1 m/s. The true velocity falls in the range $[6V_{\max},\ 7V_{\max}]$, indicating an ambiguity index of $\zeta = 3$. Fig. 10 shows the comparison of the angle spectrum under no compensation, incorrect compensation, and correct ambiguity compensation, highlighting the importance of accurate velocity estimation and correct phase compensation in ensuring image focusing and precision of the reconstruction. As shown in Fig. 11, compared with other index estimation methods, the proposed MHPC algorithm yields the most distinct response at the correct index of $\zeta = 3$ and maximizes contrast with adjacent indices to maintain robustness under noise.

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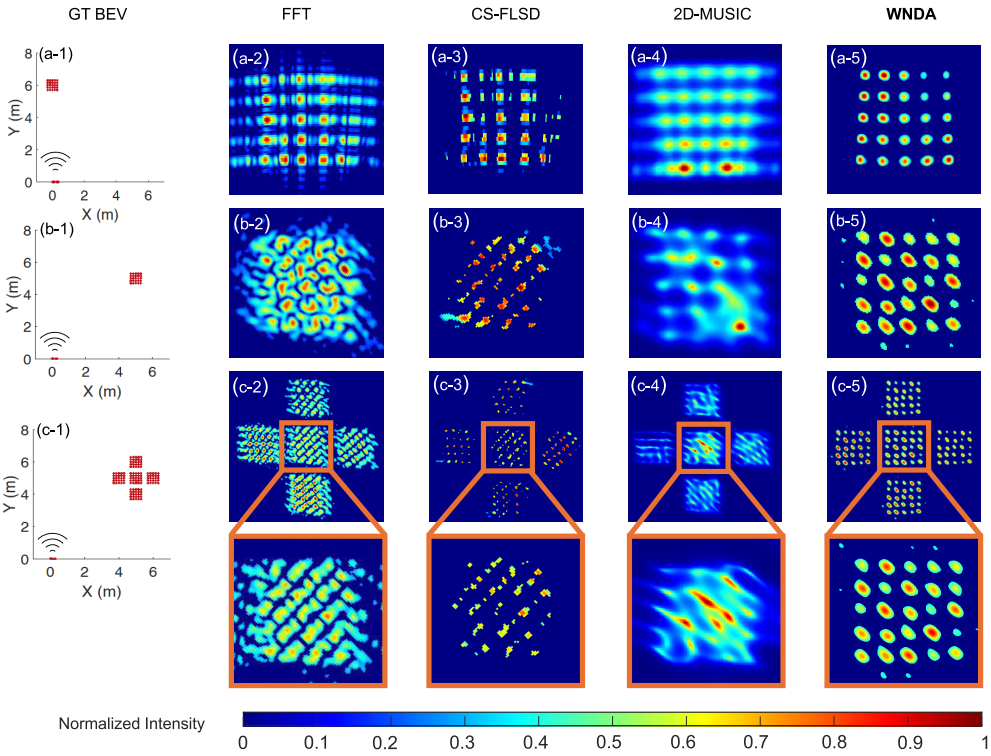


Fig. 12. Comparison of imaging results for point target scenes. Row 1: a 5×5 grid of targets centered at [0, 5] m. Row 2: a 5×5 grid centered at [5, 5] m. Row 3: a cross-shaped $5 \times 5 \times 5$ target layout centered at [5, 5] m. Each row includes, from left to right: GT bird's-eye view (BEV), FFT, CS-FLSD [26], 2D-MUSIC [25], and the proposed WNDA method.

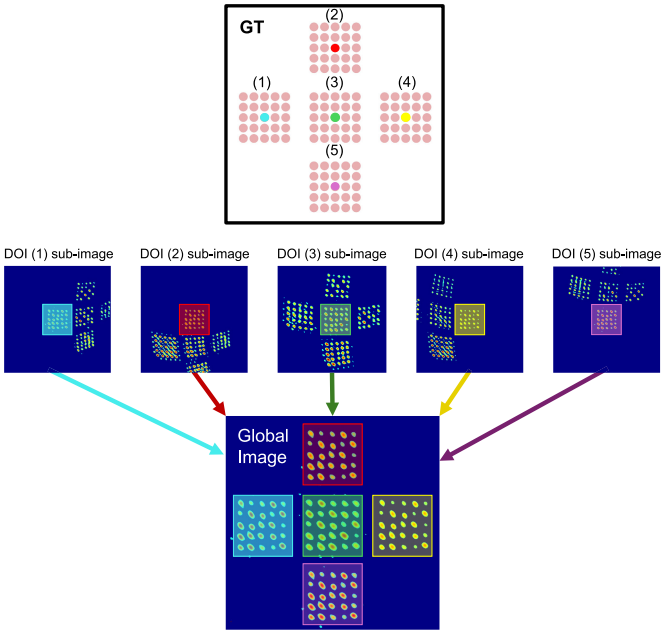


Fig. 13. Illustration of the WNDA imaging flow, demonstrating the idea presented in Section III-C2.

C. Validation of WNDA Imaging

In this part, the WNDA imaging method is evaluated against several baseline approaches, including traditional FFT, the superresolution 2D-MUSIC algorithm [25], and the SOTA fast Lq sparsity-driven (FLSD) [26], using both simulated and real-measured data. In addition, imaging with a nonuniform MIMO array and the 3-D imaging capability were also validated. Finally, the computational complexity is analyzed.

1) Evaluation Metrics: Chamfer distance (*CD*) is a supervised metric that measures the geometric similarity between the reconstructed scatterer coordinates and the GT target distribution. The *CD* is defined as

$$CD\left(\mathcal{P}_{\text{rec}}, \mathcal{P}_{\text{gt}}\right) = \frac{1}{|\mathcal{P}_{\text{rec}}|} \sum_{\mathbf{r'}_i \in \mathcal{P}_{\text{rec}}} \min_{\mathbf{r'}_j \in \mathcal{P}_{\text{gt}}} \left|\mathbf{r'}_i - \mathbf{r'}_j\right|^2 + \frac{1}{|\mathcal{P}_{\text{gt}}|} \sum_{\mathbf{r'}_j \in \mathcal{P}_{\text{gt}}} \min_{\mathbf{r'}_i \in \mathcal{P}_{\text{rec}}} \left|\mathbf{r'}_j - \mathbf{r'}_i\right|^2 \tag{26}$$

where $\mathcal{P}_{\text{rec}}$ and $\mathcal{P}_{\text{gt}}$ denote the reconstructed and GT point sets, respectively. A lower *CD* value indicates a higher level of alignment between the reconstructed and true target distributions. Furthermore, image entropy (IE) is used as an unsupervised metric to quantify the focusing quality of the reconstructed image $\mathbf{X}$, which is defined as

$$\text{IE} = -\sum_{i,j} \mathbf{X}(i,j) \log \mathbf{X}(i,j) \tag{27}$$

where $\mathbf{X}(i, j)$ denotes the normalized pixel value at position (i, j), and the matrix $\mathbf{X}$ satisfies $\sum_{i,j} \mathbf{X}(i, j) = 1$. A lower IE value indicates stronger energy concentration and better image focus.

2) Simulated Data: The simulated data is generated using a 64-Tx–96-Rx uniform linear array (ULA) MIMO radar, with waveform details listed in Table I. To comprehensively validate

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TABLE III
COMPARISON OF CD AND IE UNDER DIFFERENT NOISE LEVELS

SNR (dB)	FFT		CS-FLSD		2D-MUSIC		**WNDA**	
	CD	IE	*CD*	IE	*CD*	IE	*CD*	IE
−15	0.0370	6.3500	0.0202	**3.909**	0.3510	4.1212	**0.0115**	4.721
−10	0.0349	6.3250	0.0195	**3.701**	0.3020	3.9356	**0.0080**	4.594
−5	0.0317	6.1182	0.0192	**2.932**	0.0267	3.6488	**0.0054**	4.542
0	0.0240	5.6066	0.0168	**2.928**	0.0244	3.4321	**0.0026**	4.536
5	0.0194	5.0580	0.0168	**2.921**	0.0207	3.4095	**0.0026**	4.273
10	0.0184	4.6173	0.0157	**2.919**	0.0192	3.4364	**0.0014**	4.070

the performance of the proposed algorithm, four scenarios are tested: point targets, extended objects, nonuniform MIMO array imaging, and 3-D reconstruction capability.

Point Targets: Three representative scenarios are designed under various spatial conditions. First, a 5×5 point target grid is placed at a range of 5 m, centered at coordinates $[0, 5]$ m, with a uniform spacing of 0.15 m. Unless otherwise specified, the $[x, y]$ coordinate format is used throughout this article. In the second scenario, the azimuth angle of the target array is increased to 45° by shifting the center of the grid to $[5, 5]$ m while maintaining the same spacing to test imaging performance at the squint angle. Finally, to assess the performance for extended targets, five such 5×5 grids are centered at $[5, 5]$ m to simulate a larger complex scene. Imaging results for all scenarios using the proposed method and baseline approaches are shown in Fig. 12. The proposed method exhibits consistently lower reconstruction error and maintains high structural fidelity even under large azimuth angles, clearly imaging all individual scatterers. As shown in Fig. 12(c-5), the image is formed by integrating five DOI-wise subimages to produce a distortion-free global image. The overall processing flow is illustrated in Fig. 13.

For numerical analysis, 100 randomly distributed point targets are placed within a square region defined by $x, y \in [4.5, 5.5]$ m. Additive white Gaussian noise (AWGN) is added to simulate varying noise conditions, resulting in SNR levels ranging from −15 to 10 dB. For each SNR level, the images are assessed using *CD* and IE. As presented in Table III, the proposed method achieves the lowest *CD* across all SNR conditions, indicating the highest localization accuracy and structural fidelity. Although the IE of our method is not the lowest, FLSD yields the lowest values due to its strong sparsity prior, which leads to missing targets and large zero regions. MUSIC fails to resolve closely spaced targets and is highly sensitive to SNR, resulting in degraded performance. On the contrary, our method achieves a substantially better IE than FFT, indicating improved target focus and background suppression in fast imaging. In addition, we compute the cumulative distribution function (CDF) of localization error under different noise levels to quantify the proportion of reconstructed points falling within a given error threshold. As illustrated in Fig. 14, our method reconstructs nearly all targets within 0.05-m error across all SNR levels, whereas other methods achieve this accuracy for only about 60% of the points. Furthermore, Fig. 14 also shows that MUSIC remains

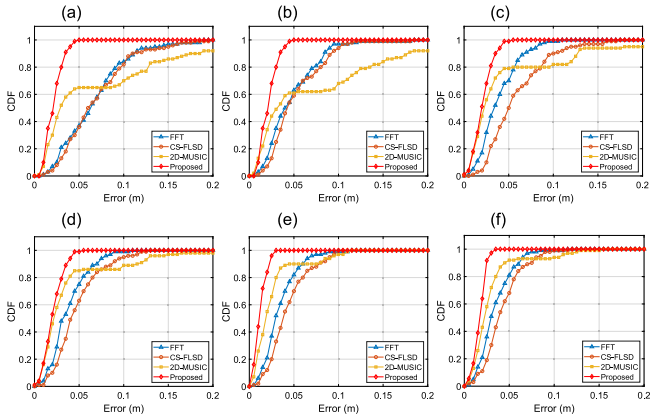


Fig. 14. CDF curve under different SNR of various baselines and the WNDA algorithm. (a) SNR:−15 dB. (b) SNR:−10 dB. (c) SNR:−5 dB. (d) SNR: 0 dB. (e) SNR:5 dB. (f) SNR: 10 dB.

highly sensitive to SNR. The FFT method achieves a higher CDF than FLSD due to its less-sparse reconstructions.

Extended Objects: To better simulate real-world large-scale environments, we utilize the real-scanned LiDAR PC from the ColoRadar dataset and assign scattering coefficients to generate synthetic radar data. Compared with isolated point targets, extended objects enable the evaluation of the mutual influence among objects and their impact on imaging quality. As shown in Fig. 15, the proposed method preserves detailed structural fidelity, making it more suitable for downstream tasks such as object recognition and environment mapping.

Nonuniform MIMO Array: As mentioned, the proposed imaging method can be deployed on arbitrarily distributed MIMO arrays, which significantly enhances radio frequency (RF) front-end design flexibility. As shown in Fig. 16, a 64-Tx-96-Rx MIMO array is randomly distributed within the range $x \in [-0.18, 0.18]$ m and $y \in [-0.1, 0.1]$ m. A 5×5 grid of target points is placed at the center location $[5, 5]$ m. The results demonstrate that our method achieves high-quality reconstruction under irregular array configuration, whereas conventional MIMO imaging approaches that rely on ULAs fail to produce valid results in such a MIMO configuration.

3-D Imaging: The above 2-D experiments are primarily designed for clearer visualization and convenient comparison. Notably, the baseline FLSD method is inherently 2-D, and applying 3-D MUSIC would require prohibitive memory resources, making it impractical on a mobile computing platform. On the contrary, the proposed method can be readily extended to 3-D imaging. As illustrated in Fig. 17, a set of $4 \times 4 \times 4$ point targets are placed at the location $[5, 5, 1]$ m, and a 128-Tx-128-Rx planar MIMO array is used for validation. While FFT-based imaging fails to accurately reconstruct the spatial positions or produce a focused image, the proposed method clearly resolves all target points and maintains precise scatterer localization. The FFT result is obtained using a $1024 \times 1024 \times 512$ 3D-FFT, whereas WNDA employs only a $256 \times 256 \times 256$ NUFFT, resulting in a memory reduction ratio of approximately 20.97 times.

3) Real-Measured Data: For real-world measurements, the TI-CAS-EVM radar was mounted on a mobile platform,

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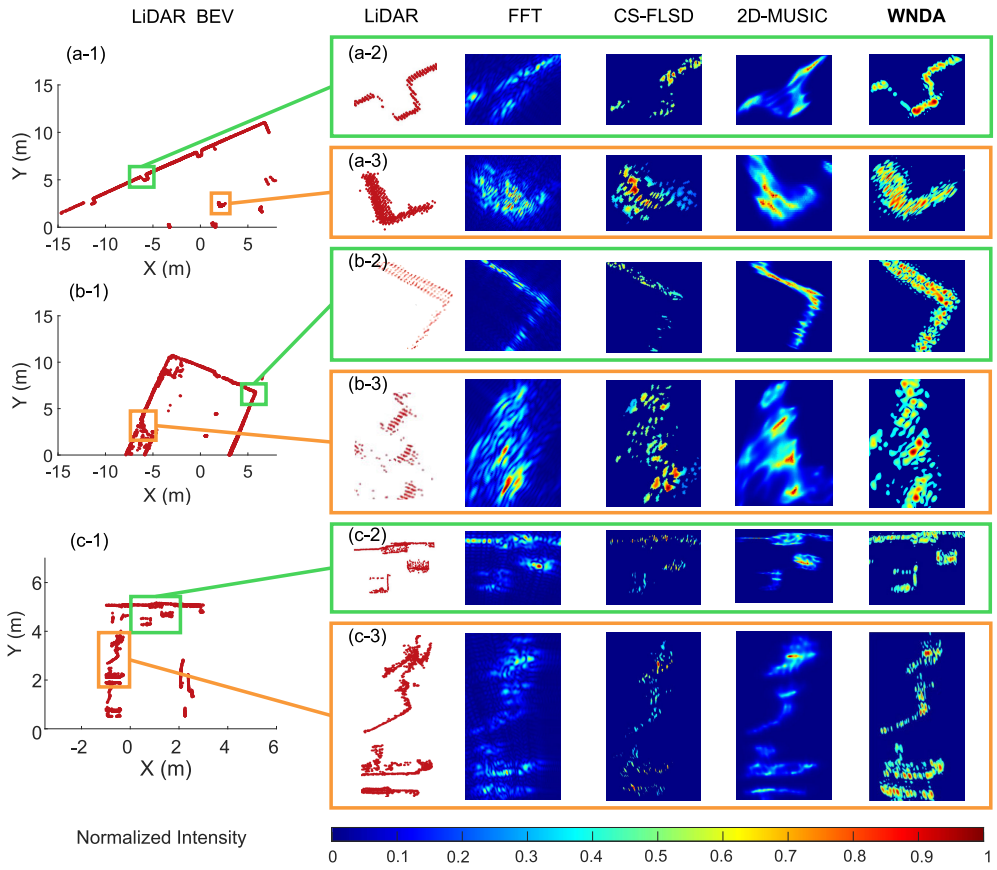


Fig. 15. Large-scale imaging results. (a-1) Outdoor scene where DOIs are set at two columns. (b-1) Garden, with two DOI subimages zoomed in on a stair and a corner, respectively. (c-1) Laboratory, with zoomed-in views of a table, printer, and server.

TABLE IV
COMPARISON OF COMPUTATIONAL COMPLEXITY AND RUNTIME UNDER VARYING SCATTERER NUMBERS

Method	**Computational Complexity**	**Time Cost (s)**		
		25 Scatterers	**50 Scatterers**	**100 Scatterers**
FFT	$\mathcal{O}\left\{N_f \tilde{K} \log_2(N_f \tilde{K})\right\}$	3.51×10^{-2}	3.61×10^{-2}	4.18×10^{-2}
CS-AMP	$\mathcal{O}\left\{N_f I_{\text{AMP}} \tilde{K}(MN)\right\}$	15.03	26.15	28.49
CS-FLSD	$\mathcal{O}\left\{N_f[I_{\text{FLSD}}(MN)^2 + \tilde{K}\log_2(\tilde{K}) + (MN)\log_2(MN)]\right\}$	21.44	42.60	66.05
2D-MUSIC	$\mathcal{O}\left\{(MN)^2 N_f + (MN)^3 + \tilde{K}(MN)^2\right\}$	82.08	82.33	89.63
WNDA	$\mathcal{O}\left\{N_{\text{DOI}}[N_f MN\sigma^2 + \tilde{K}\log_2(\tilde{K})]\right\}$	1.85×10^{-1}	1.94×10^{-1}	4.17×10^{-1}

as illustrated in Fig. 18(b). The system parameters are summarized in Table I. Experiments were conducted in an indoor office corridor, as shown in Fig. 18(a), with the imaging results shown in Fig. 19. A corner scene was also imaged, as shown in Fig. 20(a), with the corresponding imaging results shown in Fig. 20(b)-(d). It is observed that the conventional FFT method suffers from defocusing, severe sidelobes, and geometric distortions. Although the CS-based method suppresses sidelobes more effectively, it results in overly sparse images, preserving only strong scatterers. On the contrary, the proposed method yields high-fidelity reconstructions of the scene geometry. Colored boxes in the figure indicate the locations and geometric structures of objects in the RGB image and their corresponding reconstructions under the baseline and proposed methods for clear visual comparison.

4) Computational Complexity: Runtime is critical for real-time deployment on mobile IoT platforms. In this part, we analyze the computational complexity of five representative methods and provide a quantitative comparison of their time costs under simulation settings with scatterer numbers of 25, 50, and 100. In particular, $\tilde{K}$ denotes the number of discretized DOA directions or image pixels for 2D-MUSIC and WNDA, meaning the spatial domain is divided into multiple grid cells, each representing a potential target location. I_{AMP} and I_{FLSD} represent the iteration counts for the approximate message passing (AMP) [50] and FLSD algorithms, respectively; σ denotes the gridding kernel width, and N_{DOI} denotes the

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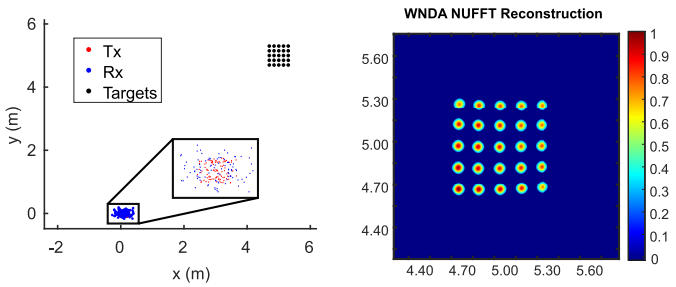


Fig. 16. Visualization of a random MIMO antenna configuration and its corresponding radar image. (a) GT points layout and MIMO configuration. (b) Simulated radar image.

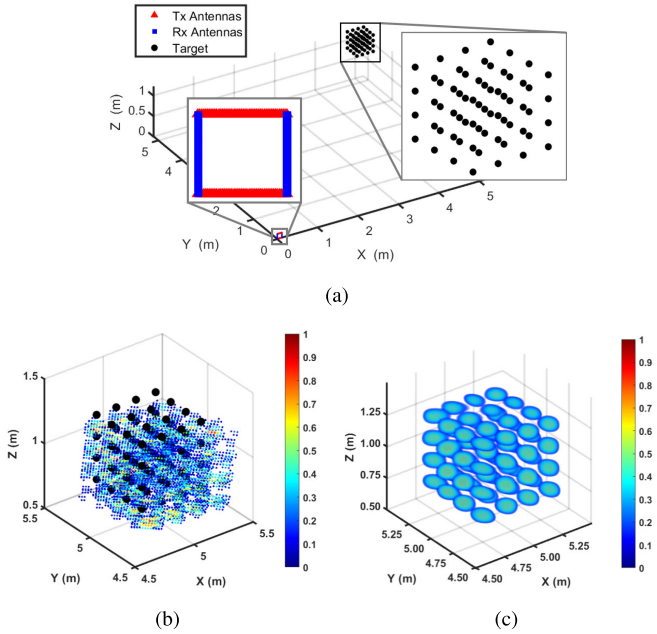


Fig. 17. (a) 3-D imaging scene and planar MIMO array configuration. (b) FFT-based imaging result. (c) WNDA imaging result.

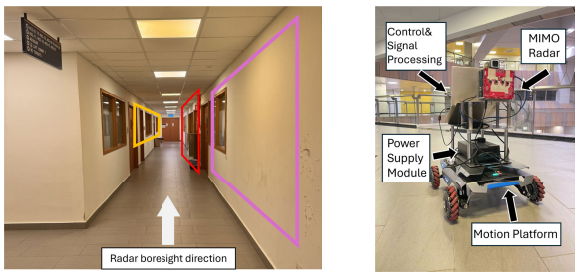


Fig. 18. Real-scene test environments. (a) Corridor imaging scene. (b) Mobile platform with MIMO radar.

number of DOIs. According to the computational complexity, as summarized in Table IV, FLSD and 2D-MUSIC exhibit quadratic and cubic growth with respect to the number of MIMO array elements, respectively, making them unsuitable for massive MIMO imaging. Although AMP is computationally more lightweight than FLSD and 2D-MUSIC, it

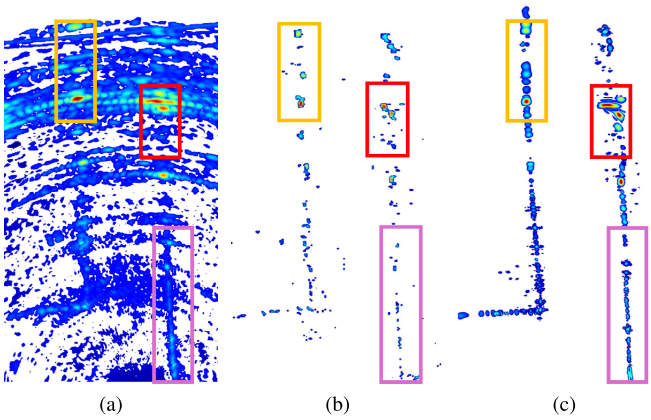


Fig. 19. Comparison of imaging results for a corridor scene. (a) FFT-based imaging result. (b) CS-FLSD imaging result. (c) WNDA imaging result.

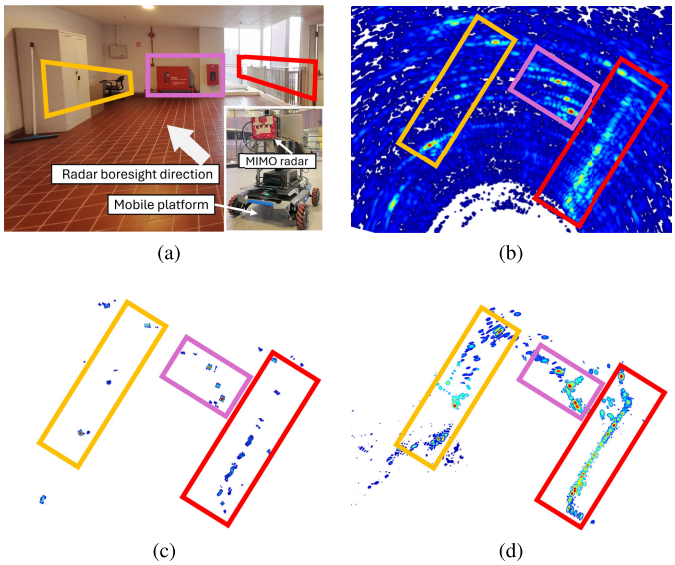


Fig. 20. Comparison of different imaging methods for the corner scene. (a) RGB image. (b) FFT-based imaging result. (c) CS-FLSD imaging result. (d) WNDA imaging result.

still requires iterative optimization, leading to longer runtime overhead.

On the contrary, the proposed WNDA algorithm employs FFT-based butterfly operations and enables parallel processing across independent DOIs, significantly improving imaging efficiency. For each DOI, only a low-dimensional NUFFT is applied, reducing errors from zero padding and avoiding the computational overhead of high-dimensional FFTs. Experimental results show that WNDA needs only an order of magnitude higher runtime than conventional FFT, while substantially outperforming iterative and subspace-based methods. These baseline methods require memory-intensive eigen-decomposition and spectral peak searches, making them impractical for real-time IoT applications.

In summary, WNDA offers a favorable balance between computational cost and imaging performance. It introduces only a modest time consumption compared to FFT while achieving significantly higher reconstruction fidelity and

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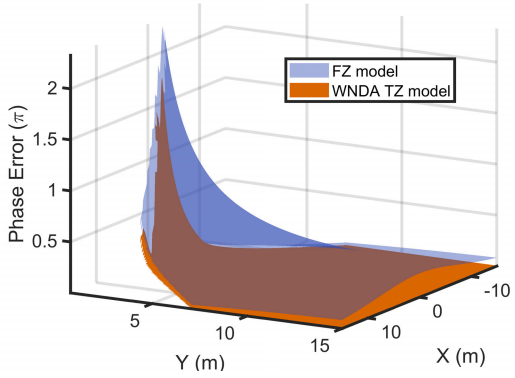


Fig. 21. Phase error comparison of different imaging models.

much lower complexity and runtime than other baseline techniques.

V. Analysis and Discussion

A. Model Accuracy Analysis

To investigate the impact of propagation models on reconstruction fidelity, we consider a ULA comprising 86 elements, the same as the horizontal virtual array layout of the TI-CAS-EVM. According to the waveform configuration, as presented in Table I, the radar's maximum unambiguous range is approximately 15.18 m. An FOV domain, with $x \in [-15, 15]$ m, $y \in [0.8, 15]$ m, and an observable angle $\pm 65°$, is discretized using a uniform square grid with a side length of 0.01 m. For each grid point, the phase delay and propagation distance from the target to each antenna element are computed under three models: the NZ model using exact Euclidean distances, the FZ model assuming planar wavefronts, and the proposed TZ model. In array imaging, the critical factor is the relative phase differences across antenna elements. With the NZ model as the reference, phase errors of the FZ and TZ models are computed. To ensure generality and represent practical subscene imaging with the TZ model, the target is offset by $[0.3, 0.3]$ m from the DOI center, implying an effective subimage size of approximately 1×1 m. As shown in Fig. 21, the TZ model provides lower phase error than the FZ model over the entire FOV, which directly contributes to improved image quality. The cross sections of Fig. 21 are shown in Fig. 22(a)–(c).

To further visualize the differences, we compute the phase difference between adjacent antenna elements within the ULA for a target located at $[5, 5]$ m. As illustrated in Fig. 22(d), the FZ model yields a constant phase difference due to the planar wavefront assumption, resulting in a flat differential profile. On the contrary, the TZ model closely matches the NZ model GT, with spatially varying phase differences, which leads to superior reconstruction accuracy. The same conclusions also hold for other nonuniform MIMO array configurations.

B. Data Utilization Efficiency Analysis

A MIMO radar configuration with a 128-Tx-128-Rx ULA is considered. The horizontal spacing between the Tx array and Rx array is 0.1 m, and the vertical spacing is 0.05 m. The

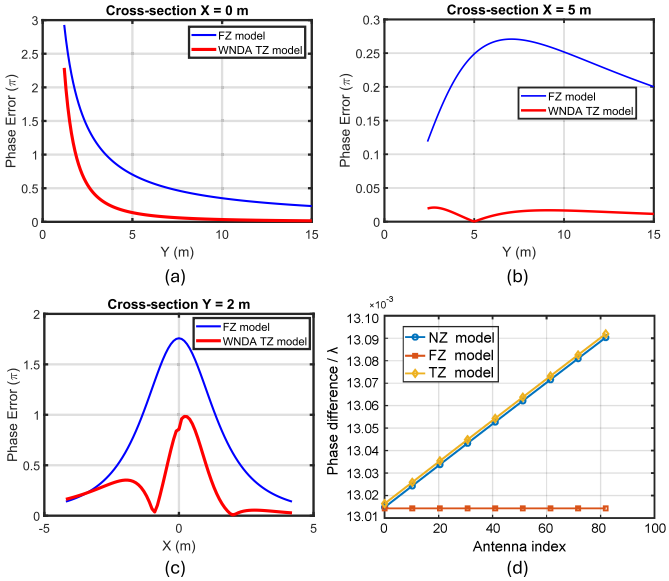


Fig. 22. Comparison of phase error cross sections under different imaging models and interelement phase differences. (a) Phase error along $x = 0$ m. (b) Phase error along $x = 5$ m. (c) Phase error along $y = 2$ m. (d) Phase difference of adjacent antenna elements.

corresponding equivalent single-input multiple-output (SIMO) virtual array contains 192 elements [39]; however, a substantial number of virtual channels are spatially overlapped. Although the full MIMO setup yields 16 384 (128^2) virtual channels, a large portion is redundant because of spatial overlap, leading to low effective data utilization. On the contrary, when the MIMO measurements are mapped to the wavenumber domain of a specific DOI (e.g., centered at 1.6 m from the origin), only six samples exhibit overlap. The remaining 16 378 unique samples contribute to the image reconstruction, significantly improving data utilization. Although this example reflects one specific case, similar phenomena are observed in commercial radar modules such as the TI-CAS-EVM. Even with a carefully engineered antenna layout, only 86 out of 144 horizontal virtual array elements are spatially unique. On the contrary, our proposed wavenumber-domain method retains all samples without overlap and utilizes them effectively, contributing to its superior imaging fidelity compared to conventional virtual array-based methods.

VI. Conclusion

This article presented a mmWave MIMO radar imaging method that integrates wave physics modeling with deep learning for high-fidelity and efficient environment reconstruction on mobile IoT platforms. Experimental results demonstrated that the proposed method improves target detection accuracy by 23.5% and enhances reconstruction quality with an 84.5% reduction in the *CD* compared with the baseline approach, while still maintaining subsecond imaging speed. The proposed method offers a novel perspective for environment perception and reconstruction on resource-limited mobile IoT platforms for V2X and smart city applications. Future work will focus on extending the framework to multisensor cross-modal fusion for robust perception and semantic reconstruction in complex environments.

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APPENDIX A
TAYLOR EXPANSION OF (3)

To approximate the round-trip propagation path in (3), the time-varying distance from the mth Tx to the qth scatterer is given as follows:

$$R_{m,q}(t) = \Big[\left(x_{m,q} + v_q t \cos\theta_{vq} \sin\varphi_{vq}\right)^2 + \left(y_{m,q} + v_q t \cos\theta_{vq} \cos\varphi_{vq}\right)^2 + \left(z_{m,q} + v_q t \sin\theta_{vq}\right)^2 \Big]^{1/2} \tag{28}$$

where φ_{vq} and θ_{vq} denote the azimuth and elevation angles of the scatterer velocity, with $v_q = \|\mathbf{v}_q\|$. The second-order Taylor expansion of $R_{m,q}(t)$ at $t = 0$ is given by

$$R_{m,q}(t) \approx R_{m,q}(0) + \left.\frac{dR_{m,q}(t)}{dt}\right|_{t=0} t + \frac{1}{2}\left.\frac{d^2R_{m,q}(t)}{dt^2}\right|_{t=0} t^2. \tag{29}$$

The first and second derivatives at $t = 0$ are given as

$$\left.\frac{dR_{m,q}(t)}{dt}\right|_{t=0} = \frac{v_q C_m}{R_{m,q}(0)}$$
$$\left.\frac{d^2R_{m,q}(t)}{dt^2}\right|_{t=0} = \frac{v_q^2\left(R_{m,q}(0)^2 - C_m^2\right)}{R_{m,q}(0)^3} \tag{30}$$

where the directional projection term is defined as

$$C_m = x_{m,q}\cos\theta_{vq}\sin\varphi_{vq} + y_{m,q}\cos\theta_{vq}\cos\varphi_{vq} + z_{m,q}\sin\theta_{vq}.$$

Substituting (30) into (29), we obtain

$$R_{m,q}(t) \approx R_{m,q}(0) + \frac{v_q C_m}{R_{m,q}(0)} t + \frac{v_q^2\left(R_{m,q}(0)^2 - C_m^2\right)}{2R_{m,q}(0)^3} t^2 \approx R_{m,q}(0) + v_{r,m,q} t \tag{31}$$

where $v_{r,m,q} = v_q C_m / R_{m,q}(0)$ denotes the radial velocity of the scatterer with respect to the transmitter. Then the total round-trip distance from transmitter m to scatterer q and back to receiver n can be approximated by

$$R_{m,n,q}(t) = R_{m,q}(t) + R_{n,q}(t) \approx R_{m,q}(0) + v_{r,m,q} t + R_{n,q}(0) + v_{r,n,q} t = R_{m,n,q} + v_{r,m,n,q} t. \tag{32}$$

The second-order term neglected in (31) gives the leading approximation error of the linear model. The Tx-path error is

$$e_{m,q}^{\mathrm{Tx}}(t) \approx \frac{v_q^2\left(R_{m,q}^2(0) - C_m^2\right)}{2R_{m,q}^3(0)} t^2 = \frac{v_{\perp,m,q}^2}{2R_{m,q}(0)} t^2 \tag{33}$$

where $v_{\perp,m,q}^2 = v_q^2 - v_{r,m,q}^2$, and the same form holds for the Rx-path error. To ensure a focused image, two-way path error over one frame time is required to be less than $\lambda_0/8$, which bounds the range and velocity as

$$\frac{v_{\perp,m,q}^2}{R_{m,q}(0)} + \frac{v_{\perp,n,q}^2}{R_{n,q}(0)} \le \frac{\lambda_0}{4T_s^2}. \tag{34}$$

APPENDIX B
DERIVATION OF WAVENUMBER-DOMAIN APPROXIMATION

If the conditions $|\mathbf{r}_T| \gg |\mathbf{r}'|$, $|\mathbf{r}_R| \gg |\mathbf{r}'|$, and $K|\mathbf{r} - \mathbf{r}'| \gg 1$ are satisfied, the scalar distance between the antenna at $\mathbf{r}$ and a scatterer at $\mathbf{r}'$ can be expressed as

$$|\mathbf{r} - \mathbf{r}'| = \sqrt{R^2 - 2\mathbf{r}\cdot\mathbf{r}' + |\mathbf{r}'|^2} = R\sqrt{1 - \frac{2}{R}\hat{\mathbf{r}}\cdot\mathbf{r}' + \frac{|\mathbf{r}'|^2}{R^2}} \tag{35}$$

where $R = |\mathbf{r}|$ and $\hat{\mathbf{r}} = \mathbf{r}/|\mathbf{r}|$. Defining

$$\delta = \frac{2}{R}\hat{\mathbf{r}}\cdot\mathbf{r}' - \frac{|\mathbf{r}'|^2}{R^2} \tag{36}$$

and applying the Taylor series expansion to $\sqrt{1-\delta}$, we obtain

$$\sqrt{1-\delta} = 1 - \frac{1}{2}\delta - \frac{1}{8}\delta^2 - \frac{1}{16}\delta^3 + \cdots \tag{37}$$

Therefore, the scalar distance becomes

$$\left|\mathbf{r} - \mathbf{r}'\right| = R\left(1 - \frac{1}{2}\left(\frac{2}{R}\hat{\mathbf{r}}\cdot\mathbf{r}' - \frac{|\mathbf{r}'|^2}{R^2}\right) - \frac{1}{8}\left(\frac{2}{R}\hat{\mathbf{r}}\cdot\mathbf{r}' - \frac{|\mathbf{r}'|^2}{R^2}\right)^2 + \cdots\right). \tag{38}$$

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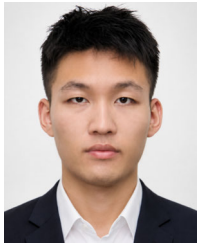
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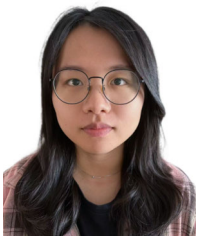
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